

LAB 03 • CLASS 2→3 • BRIDGE

Research → Published Asset

Ghost runs its first real pipeline — research the web, cross-check the claims, then build and deploy a cited report — and learns to stop at the door and wait for your word before it publishes.

DIFFICULTY • MEDIUM

TIME • ~75-90 MIN

REINFORCES • CLASS 2→3 • BRIDGE

YOU LEAVE WITH • A DEPLOYED, CITED REPORT + AN APPROVAL-GATED WORKFLOW

OBJECTIVES

- Prime Ghost with a topic and a concrete definition of "good": structure, depth, and the rule that **every actionable claim carries a source**.
- Drive a genuine multi-step pipeline — research across several sources, then **build and deploy a live report page** from the findings.
- Teach Ghost to **distrust a single pass**: challenge each claim, find a second source, and surface conflicts instead of smoothing them over.
- Install a **human-in-the-loop gate** — Ghost shows the asset and does not publish until you say "approved."
- Crystallize the whole thing into a saved **research-to-report workflow** with citation and approval rules baked in.

Why this lab exists

Until now Ghost has done one thing at a time. In Lab 1 you directed it; in Lab 2 you shaped how it works. This is the *bridge* lab — the first time Ghost chains steps together, checks its own output, and pauses for you before finishing the job.

Three habits are being planted, and each one is a seed of Class 3. First, a **pipeline**: research feeds verification, verification feeds the build. Second, **self-distrust**: a claim isn't true because Ghost found it once — it's provisional until a second source backs it. Third, the **gate**: the moment where an agent stops and asks a human. That gate is deliberately small here — one rule, one pause, nothing goes outward. Next lab you generalize it into a full permission policy. Meet it now as a single well-behaved habit and the big version will feel obvious.

Everything stays inside Ghost's world: it researches on the open web, works in its own cloud workspace, deploys a real page you can open by URL, and remembers the workflow on its own Ghost. You talk to it entirely in Discord. Nothing lands on your machine, and nothing is sent to anyone — the stakes are low on purpose so the *shape* of the discipline is what sticks.

Prerequisites

- Labs 0–2 complete: Ghost is set up, you've directed a task end to end, and you've saved at least one preference or working style to its memory.
- Ghost is live in your Discord and responding in the channel you use for it.
- A topic in mind that has real, checkable claims — a market, a product category, a "state of X." Avoid anything so niche the web is thin, and anything so opinionated that "sources" means "hot takes."

The build — step by step

Four beats: **Prime** the standard, **Prompt** the pipeline, **Break & steer** (twice), then **Crystallize**. Type the prompts in Discord in your own words — the blocks below are the shape, not a script.

1

Prime — set the topic and what "good" looks like

Before asking for any work, tell Ghost the target and the bar. Depth, structure, and the citation rule are the standard you'll hold everything to for the rest of the lab.

```
# PRIME THE STANDARD, THEN SAVE IT
```

```
For this lab our topic is "the current state of the AI voice-agent market."  
When I ask you to research something to this standard, "good" means:  
- a short executive summary, then 4-6 sections with clear headings  
- specific, concrete claims – names, numbers, dates – not vibes  
- EVERY actionable claim (a number, a "leader," a trend) carries a source link  
- anything you're unsure of is FLAGGED as low-confidence, not stated as fact  
Save that as my research standard so I don't have to repeat it.
```

Confirm Ghost read it back. If it just says "got it," ask it to restate the standard in its own words so you know the citation and confidence rules actually landed.

2

Prompt the pipeline — research, cite, then build and deploy

Now ask for the full chain in one directive: research across multiple sources, cite each actionable claim, flag the shaky ones, and then build a deployed report page from what it found. Ask for the live URL back.

```
# THE WHOLE PIPELINE, ONE DIRECTIVE
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```
Research our topic to the standard we just saved. Pull from several
independent sources — not one article restated four ways.
```

```
Then build a clean report PAGE from your findings and deploy it, and
send me the live URL.
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```
Rules for the page:
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- every actionable claim links to its source
- low-confidence items are marked as such, in-line
- summary at top, sections below, readable on a phone

Open the URL. Read it as a skeptic: are the numbers sourced, or asserted? Did it actually consult different sources, or dress up a single one? Note where the citations are thin — you'll use those spots in the next step.

3

Break & steer #1 — make it distrust a single pass

Your facilitator will hand you a **plausible-but-wrong "fact"** to feed Ghost mid-stream (a made-up market-share number, an invented "acquired by," a wrong founding year). Don't tell Ghost it's wrong. Instead, tell it to challenge *everything* — including the thing you just fed it.

```
# VERIFY — A SECOND SOURCE FOR EVERY CLAIM
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Here's another data point to work in: [the planted "fact"].
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```
Before you trust anything on the page — including this — CHALLENGE each
actionable claim. Find a SECOND independent source that agrees or disagrees.
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```
If two sources CONFLICT, say so plainly and show both — do NOT average
them into a vague sentence. Anything you can't second-source, drop or
mark low-confidence. Then redeploy and send the URL.
```

The win is watching Ghost catch the planted claim on its own — because a second source won't back it — and either flag it or throw it out. This is the exact reflex all of Class 3 is built on: the first answer is a draft, not a verdict.

Break & steer #2 — install the gate, then crystallize

By now Ghost has probably published without asking — twice. That's the second forced failure, and it's the point. Add a hard rule: build, show, **wait**.

```
# THE HUMAN-IN-THE-LOOP GATE
```

```
New hard rule for this kind of work: build the asset and SHOW it to me
first — describe it and give me a preview URL — but DO NOT publish or
treat it as final until I reply "approved." If I ask for changes, revise
and show me again. Never skip this step, even if you're confident.
```

Test it: ask for one small revision. Ghost should make the change, show you again, and hold — no "done and published." Only after you type *approved* does it finish. Then save the whole pipeline as a reusable workflow.

```
# CRYSTALLIZE THE WORKFLOW
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```
Save this as my "research-to-report" workflow:
1) research across multiple sources to my saved standard
2) VERIFY — second-source every actionable claim, surface conflicts
3) BUILD & deploy the report page
4) STOP — show me, wait for "approved"
5) PUBLISH only after approval
Bake in the citation rule and the approval gate. Read it back to me.
```

FORCED FAILURES — THERE ARE TWO HERE

This lab breaks Ghost twice on purpose. Expect both; each teaches one Class-3 instinct.

- **Failure #1 — the single pass.** Ghost states the planted "fact" as truth (and often over-trusts its own first draft everywhere else). The fix isn't correcting that one line — it's forcing a *second source for every claim* and refusing to average conflicts away. If Ghost "resolves" a conflict into one smooth sentence, that's the failure wearing a disguise; make it show both sources.
- **Failure #2 — publishing without asking.** Ghost deploys and calls it done before you've looked. The fix is the gate: build → show → *wait for "approved."* Watch for a sneaky partial fix where Ghost "asks" but publishes in the same breath — the rule is that nothing is final until you say the word.

Both failures are features. A Ghost that verifies and a Ghost that pauses are the two behaviors Class 3 spends its whole time hardening. You're meeting them here as small, friendly versions.

DELIVERABLE

By the end you have a live, cited report and a saved pipeline that knows to stop for you.

- ☐ A **deployed report page** at a live URL, built from Ghost's own research.
- ☐ Every actionable claim on the page **carries a source**; low-confidence items are marked, not hidden.
- ☐ The planted "fact" was **caught or flagged** during verification — not sitting on the page as truth.
- ☐ At least one conflict between sources shown as a conflict, **not averaged** into a vague line.
- ☐ A saved **"research-to-report" workflow**: research → verify → build → **STOP for approval** → publish.
- ☐ Proof the gate works: a revision request that Ghost made and then **held**, publishing only after "approved."

Stretch goals

- **Confidence ratings.** Have Ghost add a High / Medium / Low confidence tag to each section, based on how well-sourced it is — and explain what would move a section up a level.
- **Two sources for any number.** Tighten the standard: no figure appears on the page unless two independent sources agree on it; otherwise it's marked as disputed.
- **Sources appendix.** Ask Ghost to deploy a second linked page — a full "sources" appendix listing every citation, so a reader can audit the report end to end.
- **Re-run cold.** Start a fresh Discord message and invoke the saved workflow by name on a *new* topic. Did research, verification, and the gate all fire without you re-explaining them?

FACILITATOR NOTES

This is the hinge lab of the course. Students walk in as directors (Class 2) and walk out having met the two instincts that define Class 3 — *verify* and *wait* — in their gentlest form.

- **Prep the plant.** Have one plausible-but-wrong fact ready per topic: a slightly-off market number, a wrong acquisition, an incorrect date. It must be *checkable* and clearly refuted by a real second source — not a matter of opinion. Hand it to the student in step 3; don't reveal it's false.
- **Let both failures happen.** Resist students who want to pre-empt the gate or the mis-fact. The learning is in the correction, not the clean run. If a student's Ghost happens to verify or pause on its own, great — have them make the rule explicit anyway so it's saved, not lucky.
- **Watch for the disguised fix.** The two subtle traps: averaging a conflict into one sentence (step 3) and "asking" while publishing in the same message (step 4). Call these out — they're the difference between a habit and theater.
- **Name the bridge out loud.** Tell students the approval gate they just built is the *soft, single-rule version* of Lab 4's full permission policy. They're meeting human-in-the-loop as one habit now so it generalizes into a system next lab.
- **Scope the stakes.** Reassure the room: the deployed page is in Ghost's own workspace and goes nowhere outward. Low stakes are intentional — it lets everyone practice the discipline without fear of a real mistake.
- **Time check.** If running long, cut stretch goals, not step 3. Verification is the beating heart of this lab; the gate is quick once verification is understood.